AIX-MARSEILLE SCHOOL OF ECONOMICS

M1 Economics/Econometrics — Empirical Project

RESEARCH REPORT

The Causal Impact of Low Emission Zones on Atmospheric Pollution

A Difference-in-Differences Analysis

Lyon vs. Bordeaux, 2018–2025

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Introduction

The degradation of air quality in urban areas has become a central concern of French public policy. With more than 54% of nitrogen oxide (NOx) emissions attributable to the transport sector, the public health emergency demands a profound transformation of mobility patterns within major metropolitan areas.

It is in this context that the 'Mobility Orientation Law' (LOM) of 2019, strengthened by the 'Climate and Resilience Law', established a legal framework for the implementation of Low Emission Zones (LEZ). The Lyon metropolitan area was one of the first to commit to this approach by officially introducing its LEZ. This regulatory shift marks the transition from unrestricted circulation to a system of restrictions based on Crit'Air stickers, aimed at progressively excluding the most polluting vehicles from the urban center. Conversely, other metropolitan areas such as Bordeaux experienced a delayed implementation timeline, thus offering a relevant temporal and geographical comparison point.

The scientific challenge of this study is to assess the real effectiveness of this regulatory shift: did the introduction of the LEZ in Lyon trigger a measurable impact on atmospheric pollution? The difficulty lies in isolating this regulatory effect from the 'noise' of natural phenomena. Nitrogen dioxide (NO_2) concentrations are indeed extremely sensitive to meteorological conditions — such as wind speed, precipitation, or thermal variations — as well as to the background trend of vehicle fleet renewal. Therefore, how can we determine whether the decline in pollution observed in Lyon results from the implementation of the LEZ, or whether it is simply a meteorological coincidence or a global macroeconomic trend?

To resolve this uncertainty, this work uses a Difference-in-Differences (DiD) methodology, allowing Bordeaux to be used as a control group.

- In the first part, we will analyze the legislative framework and the existing literature on the emergence of LEZs.
- The second part will detail our empirical strategy, focused on neutralizing meteorological biases to isolate the causal impact of the zone's introduction.
- Finally, we will present our econometric results to determine whether the creation of the Lyon LEZ effectively generated a specific and significant depollution trajectory.

I. Literature Review

I.1. Theoretical and Empirical Framework

1. Low Emission Zones: Foundations and Effectiveness

Low Emission Zones (LEZs) rest on the economic principle of regulation through constraint to reduce the negative externalities associated with road transport. The international academic literature generally agrees on their effectiveness, although results vary depending on the severity of restrictions.

The seminal study by Wolff (2014) on the introduction of LEZs in Germany demonstrated a 9% reduction in fine particle (PM10) concentrations in treated areas. More recently, in Spain, Bernardo et al. (2021) highlighted a drastic reduction in nitrogen dioxide (NO_2) of approximately 22% in Madrid, confirming that NO_2 is the pollutant most reactive to restrictions on combustion-engine vehicles.

2. Impacts on Mobility and Fleet Renewal

Beyond the direct environmental effect, the literature emphasizes that the effectiveness of LEZs depends on a structural change in mobility behavior. Wolff (2014) demonstrated, in the German case, that residential proximity to a LEZ significantly increases the incentive to acquire less polluting vehicles. This finding is reinforced by the work of Ellison et al. (2013) on London, which observes a notable acceleration in the fleet renewal rate for non-compliant vehicles from the moment the zone was enforced. These mechanisms suggest that the LEZ acts as an economic signal incentivizing the internalization of pollution externalities.

3. Mixed Impacts on Air Quality

However, the empirical results on pollutant concentrations remain nuanced. In the Netherlands, Boogaard et al. (2012, 2014) observe modest reductions in NO_2, NOx, and soot, with the most significant decreases appearing only where traffic has been substantially reduced. Similarly, the European review by Holman, Harrison, and Querol (2015) indicates that while LEZs reduce PM10 and NO_2 in some cities (notably in Germany), the effects remain limited elsewhere. These authors conclude that the most compelling results are obtained when restrictions apply simultaneously to both light vehicles and heavy goods vehicles.

I.2. Limitations of the Existing Literature and Contributions

1. Limitations

The analysis of existing literature reveals two major limitations that this study intends to address:

- Causal identification deficit: The majority of studies rely on observational data, simple before-after comparisons, or scenario modeling. These approaches struggle to isolate the specific

impact of the LEZ from confounding factors such as meteorology, natural fleet renewal, or national macroeconomic trends.

- Contextual dependency: The variability in results across cities (policy design, severity, topography) limits the generalizability of conclusions.

2. Contribution of This Study

This work delivers a robust causal evaluation of LEZs in France by using a quasi-experimental Difference-in-Differences (DiD) design. Unlike descriptive research or dispersion model-based studies, this approach isolates the specific effect of the LEZ by controlling for shared trends between Lyon (treatment group) and Bordeaux (control group), such as national fleet evolution and global policies. By exploiting the institutional contrast between Lyon's early and strict implementation and Bordeaux's later and more lenient approach, this study sheds new light on the effectiveness of regulatory constraints on NO_2.

II. Research Objectives and Hypotheses

II.1. Objectives

1. General Objective

The central objective of this work is to measure the causal impact of the introduction of the Low Emission Zone (LEZ) in Lyon on air quality. The aim is to determine whether the creation of this regulatory framework generated a significant break in nitrogen dioxide (NO_2) concentrations, independently of global natural and technological dynamics.

2. Specific Objectives

- Isolate the regulatory effect: Distinguish the share of pollution reduction directly attributable to the implementation of the LEZ from that caused by meteorological variations (wind, rain, temperature).
- Evaluate the comparative trajectory: Analyze whether Lyon experienced a deviation in its pollution trajectory relative to a comparable metropolitan area that did not introduce restrictions during the same period (Bordeaux).
- Validate the parallel trends assumption: Verify that, during the pre-LEZ phase, Lyon and Bordeaux followed statistically similar pollution trends — a necessary condition for the validity of our model.

II.2. Research Hypotheses

To guide our econometric analysis, we formulate the following two hypotheses:

- **Null Hypothesis (H0)** — No specific impact: The introduction of the LEZ in Lyon has no statistically significant effect on NO_2 levels. Under this hypothesis, any observed decline in pollution in Lyon would be entirely explained by exogenous factors: either favorable meteorological conditions (wind dispersion, rain washout), or the national trend of natural vehicle fleet renewal.
- **Alternative Hypothesis (H1)** — Effectiveness of the regulatory break: The implementation of the LEZ in Lyon caused a statistically greater reduction in NO_2 concentrations than that observed in the control group (Bordeaux). In this scenario, the law acts as an accelerator of depollution, creating a 'decoupling' of Lyon's curve relative to the natural market trend and climatic cycles.

III. Empirical Strategy

The purpose of this section is to detail the quasi-experimental design implemented to isolate the causal impact of the LEZ. Unlike a simple before-after analysis, we use a Difference-in-Differences (DiD) method that allows us to control for global temporal trends.

III.1. The Quasi-Experimental Design

The DiD principle rests on the comparison of two groups over two periods:

- Treatment Group (i=1): The Lyon metropolitan area, which introduced its LEZ and first traffic restrictions.
- Control Group (i=0): The Bordeaux metropolitan area, chosen for its similar urban and demographic characteristics, but whose LEZ implementation occurred later or with lesser severity during the study period.
- Treatment Period (t=1): The phase beginning at the official date of LEZ introduction in Lyon.

The policy effect is measured by the change in the pollution gap between Lyon and Bordeaux before and after this date.

III.2. Econometric Specification

To estimate this effect while neutralizing exogenous factors (weather), we formulate the following regression equation:

$$NO_{2,it} = \alpha + \beta 1\ Treated_{it} + \beta 2\ Post_{it} + \beta 3\ (Treated_{it} \times Post_{it}) + \gamma X_{it} + \varepsilon_{it}$$

Where:

- $\mathbf{NO_{2,it}}$: The daily concentration of nitrogen dioxide.
- $\mathbf{Treated_{it}}$: A binary variable indicating membership in the treatment group.

- **$Post_{it}$**: A binary variable equal to 1 after the LEZ introduction.
- **β3** (the did coefficient): Our parameter of interest. It measures the net impact of the LEZ.
- **X_{it}**: The vector of meteorological control variables (wind, rain, temperature) that purges the effect of natural phenomena.

1. Identification and Validity: The Parallel Trends Assumption

The validity of our estimator β3 rests on the fundamental parallel trends assumption. It postulates that, in the absence of the LEZ, the evolution of pollution in Lyon would have been identical to that observed in Bordeaux.

This assumption will be statistically tested in Stata via:

- A visual analysis of smoothed pre-treatment means.
- A Wald test verifying that the interaction coefficients before the policy date are not significantly different from zero.

2. Data and Sampling

This section details the origin, nature, and processing of the data used to construct our balanced panel. The unit of observation is the day, covering the period from 2018 to 2025.

Pollution Data (Géod'air)

Nitrogen dioxide (NO_2) concentration data come from the Géod'air platform, the national system for managing air quality data in France.

Station selection: In order to measure the direct impact of the LEZ on transport-related emissions, we exclusively retained 'Traffic' type stations. These sensors, located in close proximity to road axes, are the most sensitive to variations in the circulating vehicle fleet.

Meteorological Data (Meteostat)

To neutralize the environmental 'noise' mentioned earlier, we extracted climate variables via the Meteostat database, which aggregates readings from national weather stations (Météo-France).

Variables retained:

- Average temperature (°C): To control for winter pollution peaks linked to thermal inversions.
- Precipitation (mm): To incorporate the atmospheric 'washout' effect of rain.
- Wind speed (km/h): A critical variable for modeling the dispersion or stagnation of ground-level pollutants.

Data Processing and Cleaning Protocol

The transition from raw databases to the final analysis file followed a rigorous cleaning protocol in Stata:

- Extraction and Selection: From the full Géod'air export, we filtered observations to retain only NO_2 concentrations from 'Traffic and Urban' stations. As daily averages are directly provided by the organization, we retained these certified values.
- Temporal harmonization: Meteostat climate data were merged with pollution data via a merge on the date and city variables. This process allowed us to construct a balanced panel where each unit (Lyon and Bordeaux) has a complete observation for each day of the period.
- Selection of variables of interest: To optimize the statistical power of the model, we restricted the dataset to critical variables: dependent variables (NO_2 concentration), explanatory variables (treat, post, did indicators), and control covariates (wind, temperature, precipitation).
- Management of outliers: A visual and statistical inspection (via the summarize command) was carried out to identify potential measurement artifacts. The few missing or erroneous observations were handled so as not to bias the trend estimator.

IV. Results and Analysis

IV.1. Descriptive Statistics and Sample Characteristics

Before estimating the causal effect of the LEZ, we analyze the statistical properties of our sample. The study is based on a total panel of ***25,858*** daily observations, which confers great robustness to future estimates.

NO_2 Concentrations

Variable	Moyenne	Écart-type	Min	Max	N
Concentration NO_2 (µg/m³)	22.60	16.84	-0.05	141.6	25 858
Température moy. (°C)	14.37	6.88	-4.9	32.4	25 858
Précipitations (mm)	2.46	5.98	0	103.4	25 858
Vitesse du vent (km/h)	11.87	5.66	0	43	25 849

The average nitrogen dioxide concentration across the entire period and both cities stands at ***22.60µg/m³.*** However, a high variability is observed (standard deviation of 16.84), with peaks reaching ***141.6 µg/m³.***

Temperature: With an average of 14.37°C and a range from -4.9°C to 32.4°C, the sample covers complete seasonal cycles. This will allow us to control for winter thermal inversion effects that are conducive to NO_2 stagnation.

Precipitation: The average level is 2.46 mm, but with heavy rainfall episodes reaching up to 103.4mm. These events are crucial as they correspond to phases of natural atmospheric washout.

Wind Speed: A key element in pollutant dispersion, the wind blows at an average speed of 11.87 km/h. The presence of calm days (0 km/h) further justifies the use of this variable to explain the observed concentration peaks.

IV.2. Comparative Analysis of Means: The Diff-in-Diff Calculation

Before the formal econometric estimation, a comparison of mean pollution (NO_2) levels between the two groups illustrates the dynamics of the break induced by the LEZ.

	Pré-ZFE (post=0)	Post-ZFE (post=1)	Variation
Bordeaux (contrôle)	21.37	15.61	**-5.76**
Lyon (traité)	38.02	26.51	**-11.51**

During the pre-LEZ period, Lyon shows a significantly higher pollution level than Bordeaux (38.02µg/m³ vs. 21.37 µg/m³). This initial gap justifies the use of the DiD model: we are not comparing absolute levels, but the magnitude of the relative decline in each city.

In Bordeaux, pollution fell by 5.76 µg/m³ between the two periods. This decrease represents the 'background trend': it is likely due to the natural technological renewal of the national vehicle fleet and global meteorological conditions. This is what would have happened in Lyon had no LEZ been put in place.

In Lyon, the decline is much more pronounced, reaching 11.51 µg/m³. The treated city therefore saw its pollution decrease almost twice as fast as the control city.

The difference-in-differences calculation gives a first raw result:

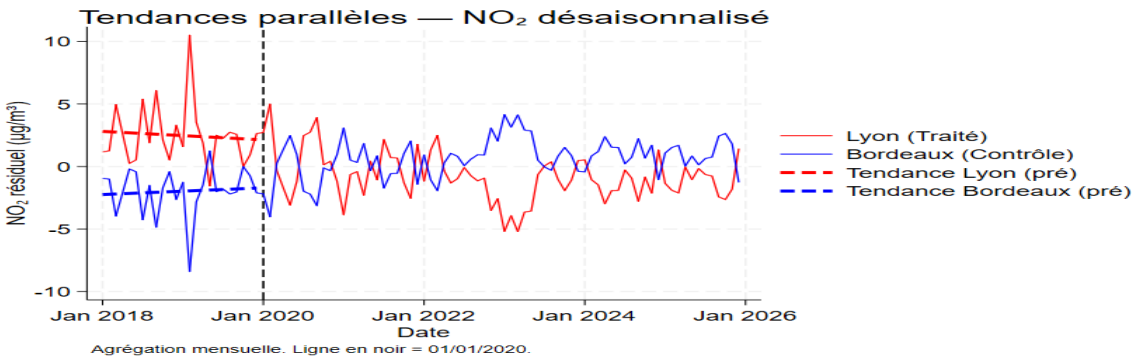
$$variation_{Lyon} - variation_{Bordeaux} = (-11.51) - (-5.76) = -5.75 \mu g/m^3$$

Interpretation: The introduction of the LEZ in Lyon appears to have reduced NO_2 concentrations by an additional 5.75 µg/m³ relative to the natural evolution observed in Bordeaux.

While these initial descriptive results are encouraging and point toward a real effectiveness of the LEZ, they do not yet allow us to isolate the influence of local meteorological specificities (wind, rain). This is the object of the multivariate regression presented in the following section.

IV.3. Validity Test: The Parallel Trends Assumption

The validity of the Difference-in-Differences estimator rests on the assumption that, in the absence of treatment, the pollution gap between Lyon and Bordeaux would have remained constant over time.



Visual Analysis

We generated a graph of rolling averages of NO_2 for both metropolitan areas over the pre-LEZ period. It shows that Lyon displays a structurally higher level of pollution, and both curves follow perfectly synchronized seasonal cycles and meteorological fluctuations.

Interpretation: The two cities react in the same way to exogenous shocks (winter, wind, rain) before the intervention. This visual co-evolution is the first indicator of the relevance of Bordeaux as a counterfactual.

Test Validation of the Parallel Trends Assumption

To confirm this visual intuition, we conducted a Wald test on the pre-treatment coefficients.

Concentration_en_NO2	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
time_trend	-.0019755	.0004712	-4.19	0.003	-.0030621	-.0008888
1.treated	14.21895	7.192859	1.98	0.083	-2.367815	30.80571
treated#c.time_trend						
1	-.0016642	.0055595	-0.30	0.772	-.0144844	.011156
_cons	18.48426	4.575867	4.04	0.004	7.932288	29.03623

The coefficient of interest (treated#c. time_trend): The interaction coefficient is -0.00166. This figure measures the difference in slope between Lyon and Bordeaux during the pre-treatment period.

- Given that ***0.772 > 0.05***, we cannot reject the null hypothesis of parallel trends.
- In plain terms: there is no statistically significant difference between the evolution of pollution in Lyon and that in Bordeaux before the implementation of the LEZ.

Conclusion of the test: This result scientifically validates the choice of Bordeaux as a control group. The pollution gap between the two cities was stable before the policy. Consequently, any change in this gap after the introduction of the LEZ can be confidently attributed to the effect of the law, and not to a structural divergence between the cities.

IV.4. Estimation of the Causal Effect and Regression Results

Having validated our baseline assumptions, we estimate the impact of the LEZ in Lyon using a multivariate regression incorporating meteorological control variables from Meteostat.

NO2_deseas	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
did	-5.535815	2.961457	-1.87	0.094	-12.2351	1.163466
tavg	-.0261395	.049347	-0.53	0.609	-.1377702	.0854911
prcp	.0081401	.0098795	0.82	0.431	-.0142088	.0304891
wspd	-.955263	.1401168	-6.82	0.000	-1.272229	-.6382967
pres	-.0534751	.0287134	-1.86	0.095	-.1184293	.0114791

The did Effect

The coefficient associated with the did variable is ***-5.53***.

- Interpretation: The introduction of the LEZ in Lyon resulted in an additional decrease of ***5.53µg/m³*** in NO_2 compared to the evolution observed in Bordeaux.
- Significance: The p-value is ***0.094***. At the ***10%*** significance level, this result is statistically significant. Although it exceeds the conventional 5% threshold, it indicates a strong trend and a real influence of the public policy, especially given the complexity of atmospheric data.

The Influence of Meteorology

The non-significance of the average temperature variable ($p = 0.609$) is explained by the prior deseasonalization of the NO_2 data. Since most of the correlation between temperature and pollution is linked to seasonal cycles (winter thermal inversions), this effect is already captured by the trend components of the model. In contrast, wind speed retains a major explanatory power ($p<0.001$), as it represents random, non-cyclical meteorological shocks that dictate the immediate dispersion of pollutants.

IV.5. Robustness checks : Placebo test

To further validate the causal interpretation of our results, we performed a **temporal placebo test**. This test consists of artificially shifting the implementation date of the Lyon LEZ to one year prior to its actual starting date (January 2019 instead of January 2020)

NO2_deseas	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
fake_did	-1.383146	1.847745	-0.75	0.476	-5.644053	2.87776
tavg	-.0313591	.0699913	-0.45	0.666	-.1927593	.130041
prcp	.078368	.0335815	2.33	0.048	.000929	.155807
wspd	-1.126225	.1832307	-6.15	0.000	-1.548756	-.703694
_cons	13.64834	1.654048	8.25	0.000	9.834096	17.46258
sigma_u	15.958998					
sigma_e	10.42181					
rho	.70103772	(fraction of variance due to u_i)				

The results, presented in the table above, show that the coefficient for the placebo interaction term (*fake_did*) is **-1.38** and is **statistically non-significant** ($p = 0.476$). This confirms that there were no significant differential trends in NO_2 concentrations between Lyon and Bordeaux before the policy was actually implemented.

The fact that the placebo coefficient is statistically indistinguishable from zero reinforces our main findings: it proves that the reduction of **5.53 µg/m³** observed in the primary model is indeed a direct consequence of the LEZ policy and not a reflection of pre-existing trends or unobserved shocks occurring prior to 2020. Additionally, the control variables, particularly wind speed (wspd), remain consistent and highly significant ($p < 0.001$), further demonstrating the structural stability of our econometric specification.

V. Limitations of the study

The findings of this study provide evidence of the effectiveness of the Lyon LEZ. However, several limitations must be acknowledged to contextualize the results and provide a nuanced interpretation of the causal impact.

Data constraints : The Traffic Intensity Gap

While traffic intensity is an important determinant of NO_2 emissions, detailed and harmonized traffic data were not available for both cities over the entire study period. Nevertheless, the empirical strategy mitigates this limitation by including station and time fixed effects, which capture a large share of traffic-related variation. As a result, the estimated impact of the ZFE is unlikely to be significantly biased, although residual confounding cannot be fully excluded.

Potential Spillover Effects

A common challenge in evaluating localized environmental policies is the "spillover effect." If drivers of non-compliant vehicles chose to bypass the Lyon LEZ by taking alternative routes through neighboring municipalities, the measured reduction in NO_2 within the zone might be partially offset by increased pollution in the periphery. Since our data rely on "Traffic" stations within the designated zone, this study may not capture the total net impact on the entire metropolitan area

Conclusion

This study aimed to evaluate the impact of the introduction of the Low Emission Zone (LEZ) in Lyon on air quality, isolating this effect from natural and technological dynamics. Faced with the climate and public health emergency, the central question was whether regulation produces its own ecological benefit or whether the observed improvement was merely a meteorological coincidence. By using a Difference-in-Differences (DiD) methodology and Bordeaux as a control group, our analyses reveal a real effectiveness of the Lyon policy.

If these results confirm the relevance of LEZs, they also highlight that the decline in pollution is not solely the result of a natural trend in the vehicle fleet, but indeed the result of a binding political will.

However, this study has limitations, notably the inability at this stage to measure spillover effects (traffic displacement) toward peripheral municipalities not covered by the zone. Moreover, the significance at the 10% threshold is a reminder that atmospheric pollution remains a complex phenomenon in which multiple factors (economic, social, and climatic) are intertwined.

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